



JC3008 Knowledge Representation

Day 1: Introduction to knowledge representation

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Day 1

Introduction to knowledge representation

- 1. Course overview**
2. Introduction to knowledge representation

Course information

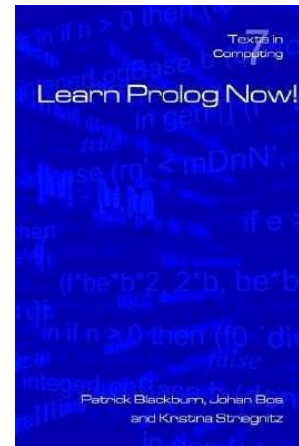
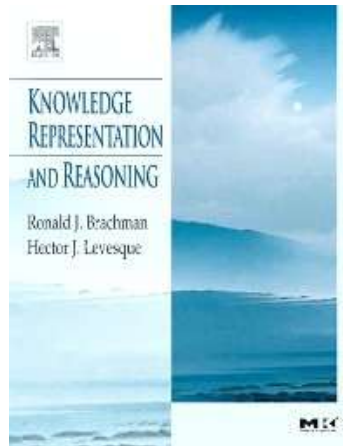
- This course will present the theory and practice of knowledge representation and reasoning
- Lectures will cover core concepts, while practical sessions will allow students to put what they have learned into practice with exercises covering theory and implementation tasks
- We will use [MyAberdeen](#) (please familiarise yourself with the system).

Plan of Lectures & Practicals

Day	Chapter(s)	Topic
1	1 & 2	Intro & FOL
2	2 & 3	Knowledge & Resolution
3	4 & 5	Horn Clauses & Logic Programming
4	Extra	Prolog I
5	Extra	Prolog II
6	14 & 15	Situation Calculus & STRIPS
7	9	Structured Description
8	9 & 16	Modern Description Logic
9	Extra	Ontology Web Language (OWL)
10	Extra	Answer Set Programming
11	–	Revision
12	–	Revision

Literature

- We cover specific chapters of *Knowledge Representation and Reasoning*, by R. Brachman and H. Levesque (see official [page](#))
- Full text available online through the university library (see [Leganto reading list](#))
- The freely available [Learn Prolog Now!](#) can also be useful



Assessment

Exam: 75% of the final course grade

(Group) Assignment: 25% of the final course grade

- The project assignment will be released 1 month after the course starts.
- It is a text-based adventure game in Prolog with planning strategy based on PDDL.
- Deadline submission is set at the end of the course.
- Marks are released 3 weeks after submission deadline.
- Written feedback is provided through the myAberdeen course site.

Learning outcomes

- Demonstrate mastery of core principles and concepts of knowledge-based systems.
- Identify the knowledge base (or lack thereof) of an AI application.
- Develop the ability to evaluate different knowledge-based systems.
- To design, implement and evaluate knowledge-intensive AI systems.
- Demonstrate an ability to perform an in-depth knowledge representation analysis of a domain and application

Ask for help if you are stuck!

- If something is unclear, then ask someone or look it up
- If you still find it hard, get in touch with teacher
- Copying something from someone else, or via the internet, will not help you learn how to do the work

It is ALWAYS better to submit a poor assignment of your work with comments about what you tried, and struggle with, than to submit someone else's work.

Activity 1

In pairs, discuss the following question (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):

“What are your expectations from this course?”

Day 1

Introduction to knowledge representation

1. Course overview
2. **Introduction to knowledge representation**

Example Systems

- We will take a look at two implemented systems
 - Cognitive Assistant (Siri, Alexa)
 - Computational Knowledge Engine (Wolfram Alpha)
- For each system, we will look at
 - What knowledge must it represent?
 - What reasoning must it do?
 - What would it take to extend it?
 - Where does it fail?
 - How is it different from a Search Engine (Google)?



Activity 2

In pairs, discuss cognitive assistants like Alexa / Siri regarding the following questions (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):



- What knowledge must it represent?
- What reasoning must it do?
- What would it take to extend it?
- Where does it fail?
- How is it different from a Search Engine (Google)?

Cognitive Assistant (SIRI)

- See Demo at: https://www.youtube.com/watch?v=MpjpVAB06O4&feature=player_embedded
- What knowledge must it represent?
 - Restaurants, movies, events, reviews, weather, ...
 - Location, tasks, web sources, ...
- What reasoning must it do?
 - Nearest location, date for tomorrow, AM vs PM, ...
- What would it take to extend it?
 - More sources, different sources, ...
- Where does it fail?
 - Unexpected response / interaction, contextual understanding, handling ambiguity, ...
- Differences from a search engine?
 - Dialog driven, task-oriented, interactions with other applications, ...

Computational Knowledge Engine

- See Demo at: [Wolfram|Alpha Examples: Nutrition \(wolframalpha.com\)](http://Wolfram|Alpha Examples: Nutrition (wolframalpha.com))
- What knowledge must it represent?
 - Foods, nutrition composition, caloric values, ...
- What reasoning must it do?
 - Mathematical computations based on portions, ...
- What would it take to extend it?
 - More data on foods, more data on nutrition composition, ...
- Where does it fail?
 - Doesn't know about recipes, how to combine foods, ...
- Differences from a search engine?
 - Focus on mathematical reasoning, data driven as opposed to document driven, ...



What can we take from those examples?

AI agent: intelligent behaviour achieved through computational means

- Requires access to knowledge
- A way of representing this knowledge
- To use the knowledge the agent will need to be able to reason about it
 - Theoretical reasoning: reasoning about things based on truth (facts) through logical and conceptual analysis.
 - Epistemic reasoning: reasoning about whether what it knows is accurate and justifiable.

What is Knowledge?

- How can we express knowledge?
 - Propositions are one way
 - “Alice knows that ...” and fill the blank with **a proposition** (e.g. Ubuntu is an operating system)
 - “John fears that Ubuntu is an operating system”
 - Same content, different attitude (Contrasting knowledge)
- Other forms of knowledge:
 - Know how, who, what, when, ...
 - Sensorimotor: typing, riding a bike
- Is knowledge always correct?
 - No, knowledge can be right/wrong, true/false
 - Knowledge can depend on time (e.g. Iphone is the most used phone in the world in Dec 2023)

What is Knowledge?

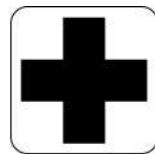
- Can discrete data be considered knowledge? For example, is the data (4, 2, 9) knowledge?
 - No, we need additional information that will turn the data into knowledge
 - In our example, we can say that (4, 2, 9) is the combination to our safe deposit box
- How about beliefs?
 - Yes, beliefs are a form a knowledge
 - Not necessarily true, and it is typically 'weaker' than facts
 - "John suspects that Ubuntu is an operating system"
- Therefore, knowledge associates some level of right/wrong or true/false to a piece of (semantic) information as viewed by one or multiple actors

Why Knowledge?

- Complex systems require complex descriptions
 - a chess playing program considering a move for the queen
 - It knows evaluation procedure using minimax returned a value of +7 for the position;
 - But it believes queen is in danger;
 - But it has a goal to keep control of the centre of the board
 - if these complex descriptions are not fully used by the system they can be misleading and lead the user to think that system knows more than it actually does.

What is Knowledge Representation?

- A symbolic way to represent knowledge / propositions believed (by some agents) and link relevant data
- Determined by a representation formalism



first aid



women



stop and give way

"John loves Mary"



the proposition that
John loves Mary

Why Representation?

- Knowledge needs to be formally represented in order to be used
- Explain and justify the behaviour of the system
 - The program did X because Y
- We can add new tasks and easily make them depend on previous knowledge
 - Enumerating objects vs painting objects
- Extend the existing behaviour by adding new beliefs
 - Assert that dragon fruits are red / pink
- Debug faulty behaviour by locating the erroneous beliefs
 - By changing the colour of sky, we change any routine that uses that information

Why Representation?

Do both of these examples have a Knowledge Base (KB)?

Example 1

```
printColour(snow) :- !, write("It's white.").
printColour(grass) :- !, write("It's green.").
printColour(sky) :- !, write("It's yellow.").
printColour(X) :- write("Beats me.).
```

Example 2

```
printColour(X) :- colour(X,Y), !,
    write("It's "), write(Y), write(".").
printColour(X) :- write("Beats me.).

colour(snow,white).
colour(sky,yellow).
colour(X,Y) :- madeof(X,Z), colour(Z,Y).
madeof(grass,vegetation).
colour(vegetation,green).
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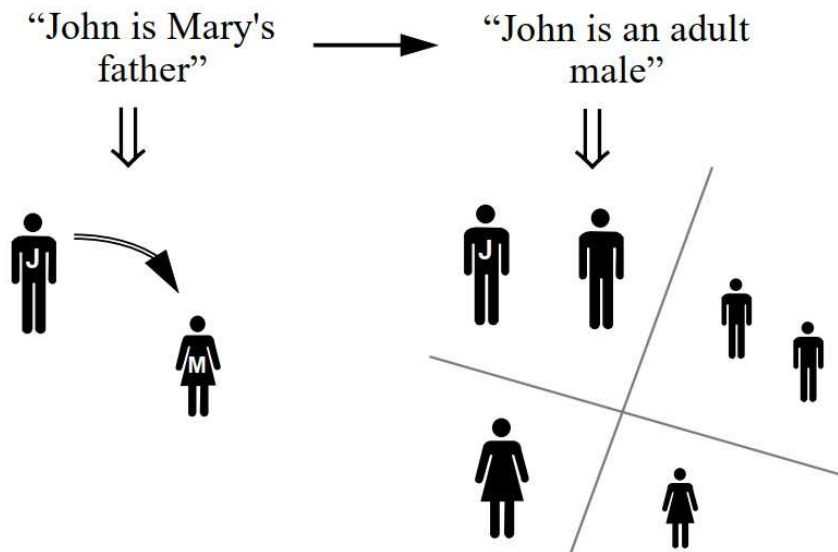
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Only Example 2 has a separate representation of symbolic structures (in KR Hypothesis representation). It is a (small) knowledge-based system.

What is Knowledge Reasoning?

- Manipulation of knowledge symbols to produce new representations
- No reason to list all existing knowledge if some of it can be inferred through reasoning

arithmetic “1011” + “10” → “1101”
 ↓ ↓ ↓
 eleven two thirteen



Why Reasoning?

1. Provides a link between knowledge and action
 2. Simple database queries are not enough to inform actions in complex systems
 3. Thus we need a mechanism to manipulate symbols in order to arrive at new ones
- Given
 - Patient X allergic to medication M
 - Anyone allergic to medication M is also allergic to medication M'
 - Reasoning helps us derive
 - Patient X is allergic to medication M'

Why Reasoning? Entailment!

Inferences and entailment (logical consequence) are usually the main reasoning mechanisms

- Entailment describes the relationship between statements that are true when one statement follows from one or more statements
 - Syntactic consequence ($\vdash$): $P_1, P_2, \dots, P_n \vdash P$ (P is provable for all P_i)
 - Semantic consequence ($\models$): $P_1, P_2, \dots, P_n \models P$ (P is true for all P_i)
- Inference describes the steps on how to move from the premises to logical consequences (i.e. the process of calculating entailments)
 - Sound: get only entailments
 - Complete: get all entailments

Logic: study of entailment relations

- Languages
- Truth conditions
- Rules of inference

Knowledge Representation & Reasoning

- KRR uses knowledge representation to describe information about the world, and then applies automated reasoning to be able to query such knowledge and to put it to use
 - Examples of knowledge representation formalisms: ontologies, formal logics, rules, semantic nets, etc.
 - Examples of automated reasoning: inference engines, theorem provers, rule-based systems, classifiers, etc.
- Two issues: existence of structures that
 - We can interpret
 - Determine how the system behaves

Is KRR useful in practice?

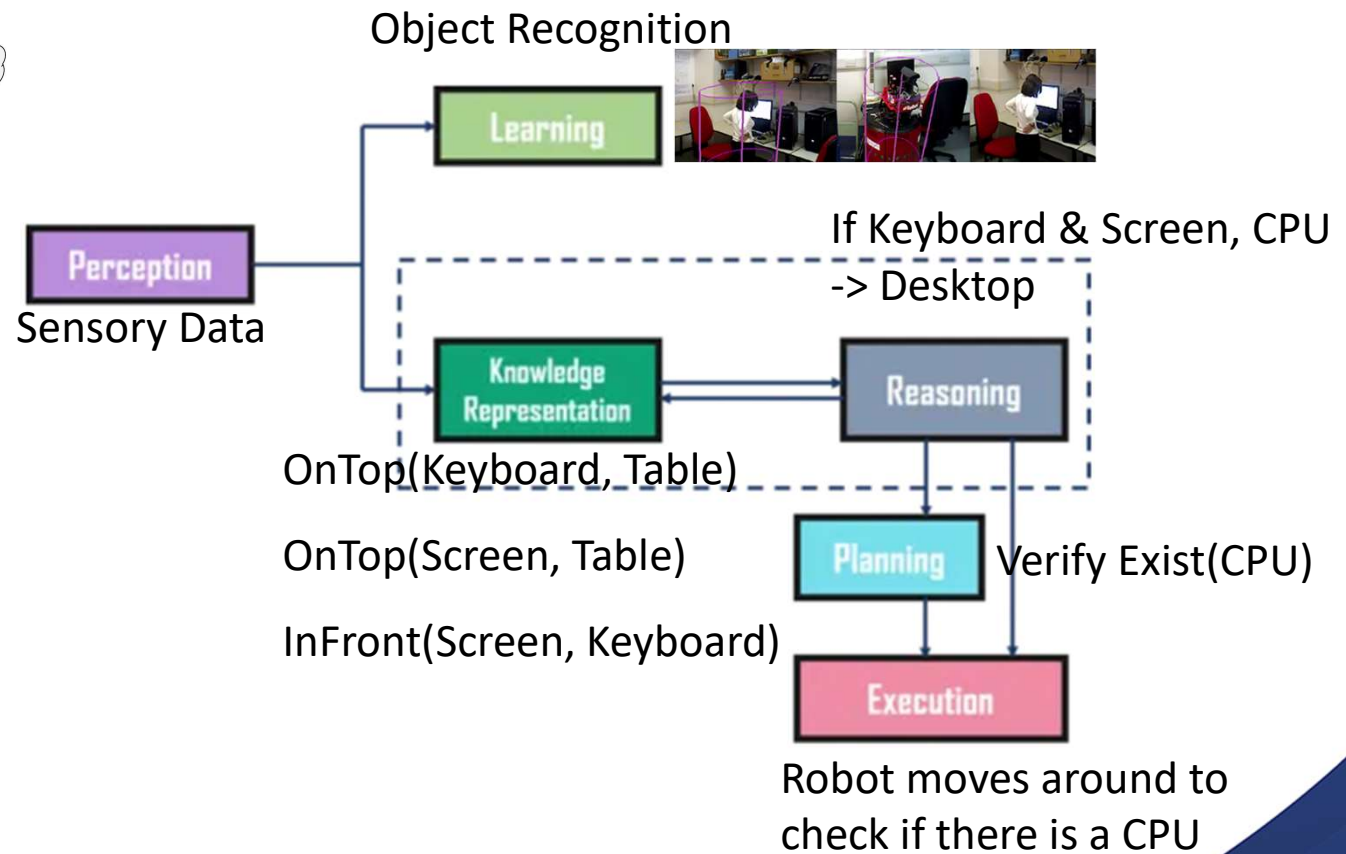
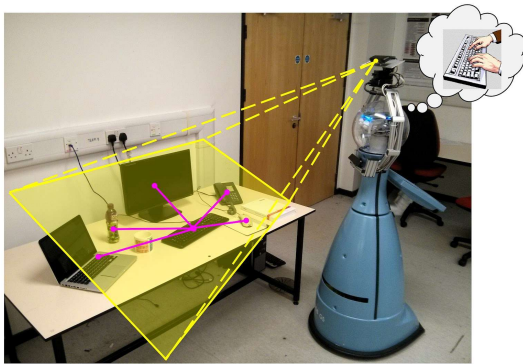
Knowledge vs data, reasoning vs reacting

- Data in databases lack semantic meaning, and thus lack reasoning
- Knowledge-based systems are most suitable for open-ended tasks
- Explainability (Explainable AI)
- Debugging
- New knowledge relationships are easy to add
- New applications are also easy to add

Cognitive (Personal) Assistants: use NLP to answer questions, use KB to make recommendations, and use web services to perform actions.



Is KRR useful in practice? Robotics



Summary

- Introduction to knowledge and how to represent and reason about it.
- Why KRR is relevant for AI
- Applications of KRR.

Until the next session:

- Install SWI-Prolog to your computer.